

Ethan Yu

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EDUCATION

Queen's University

BASc in Mechatronics and Robotics Engineering — Dean's Scholar

Expected May 2029

Kingston, Ontario

TECHNICAL SKILLS

Programming Languages: C, C++, Python, MATLAB, PostgreSQL

Embedded Systems: ARM Cortex-M4, AVR, Contiki OS, Raspberry Pi, SPI, UART, I2C, RS485, Modbus

Hardware & Tools: Oscilloscope, Multimeter, Git, KiCad, Onshape, SolidWorks

EXPERIENCE

Undergraduate Research Assistant

Queen's MEMS Laboratory

May 2026 – Aug. 2026

Kingston, Ontario

- Developed embedded C firmware for a battery-powered wireless sensor node on the ATmega256RFR2 running Contiki OS, using cooperative multitasking for sensing, networking, and power management.
- Wrote a SPI driver for a MAX31865 RTD-to-digital converter, including register-level configuration, conversion from ADC to temperature, and fault detection/clearing for a PT1000 temperature probe.
- Designed a request/response protocol over 6LoWPAN, RPL, and UDP that decodes compact codewords from a gateway to selectively collect and transmit only the requested sensor data.
- Implemented deep-sleep power management using AVR power-save mode and a wake timer to minimize energy use between transmissions, extending estimated battery life by over 17×.

EngLinks

STEM and Programming Tutor

Oct. 2025 – Present

Kingston, Ontario

- Provided one-on-one tutoring across first-year and upper-year engineering courses to 15+ university students
- Guided students through assignments and exam preparation across 3 university engineering courses

PROJECTS

Modbus Sensor

Aug. 2026 – Present

- Designed embedded C firmware for a SAMG53-based sensor node that acquires and analyzes accelerometer data over SPI, transmitting processed measurements via Modbus RTU over RS485.
- Wrote a SPI driver for a KX132-1211 3-axis accelerometer, using direct register-level PDC (DMA) control on the SPI peripheral to offload per-sample data transfer from the CPU.
- Implemented an RTC-driven acquisition scheduler collecting 9,000 samples (3,000/axis) every 2 minutes, decoupling interrupt timing from blocking acquisition to prevent peripheral stalls.

RLC Temperature Sensor Characterization

Aug. 2026 – Present

- Extended the open-source NanoVNA Saver software with Python-based CSV export and automated frequency sweeps, enabling measurements to be saved directly to a specified folder for streamlined data collection.
- Designed and tested multiple RLC PCB configurations to characterize temperature-dependent changes in bandwidth and quality factor using NanoVNA measurements.
- Analyzed frequency-response measurements across intermediate temperatures to calculate bandwidth and quality factor, identifying an RLC configuration with an absolute Q change of 12.61 over the 24°C–160°C range.

Autonomous Rover

Jan. 2026 – Apr. 2026

- Built an autonomous rover with 2 team members using ROS2, integrating LiDAR, IMU, IR sensors, and wheel encoders to achieve approximately 10 cm localization accuracy with real-time navigation.
- Designed a Raspberry Pi–Arduino control architecture, developing Python ROS2 nodes for Nav2 path planning, wheel odometry, and obstacle avoidance alongside Arduino-based PID motor control.
- Created a URDF model and RViz/Gazebo simulation environment to visualize live sensor data, TF frames, robot pose, maps, and navigation paths for system debugging, testing, and validation.